

Erdős Problem #1132

EP-1132	L1 - Tractable	Partiellement résolu	Combinatorics
---------	----------------	----------------------	---------------

Énoncé original (anglais, reproduction mot pour mot)

For $x_1, \dots, x_n \in [-1, 1]$ let $I_k(x) = \frac{\prod_{i=1}^n (x - x_i)}{\prod_{i=1}^n (x_k - x_i)}$

which are such that $I_k(x_k) = 1$ and $I_k(x_i) = 0$ for $i \neq k$.

Let $x_1, x_2, \dots \in [-1, 1]$ be an infinite sequence, and let $L_n(x) = \sum_{1 \leq k \leq n} |I_k(x)|$

where each $I_k(x)$ is defined above with respect to $x_1, \dots, x_n$.

Must there exist $x \in (-1, 1)$ such that $L_n(x) > \frac{2}{\pi} \log n - O(1)$ for infinitely many n ?

Is it true that $\limsup_{n \rightarrow \infty} \frac{L_n(x)}{\log n} \geq \frac{2}{\pi}$ for almost all $x \in (-1, 1)$?

Source et attribution. Énoncé reproduit depuis UnsolvedMath, jeu de données officiel sous licence CC BY 4.0.

Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Jeu de données: <https://huggingface.co/datasets/ulamai/UnsolvedMath>